SALES PERFORMANCE & MIS DASHBOARD

Excel-Based Sales Reporting, Data Cleaning and KPI Analysis

PROJECT REPORT
Prepared by: Sandhya Shree G M
Project Type: Data Analytics & MIS
Primary Tool: Microsoft Excel
Target Roles: MIS Executive | Data Analyst

Portfolio documentation • Excel analytics workflow • KPI reporting

Excel-Based Sales Reporting, Data Cleaning and KPI Analysis

Project Report

Prepared by: Sandhya Shree G M

Project Type: Data Analytics and Management Information System (MIS)

Primary Tool: Microsoft Excel

Target Roles: MIS Executive | Data Analyst

1. INTRODUCTION

In today's business environment, organizations need accurate and timely information to monitor sales performance, identify trends and make informed decisions. Management Information Systems (MIS) help transform raw business data into meaningful reports and dashboards.

The Sales Performance & MIS Dashboard is an Excel-based analytics project designed to demonstrate the process of organizing raw sales data, cleaning inconsistent records, calculating business performance metrics and presenting the results through a dashboard.

The project focuses on sales monitoring, profitability analysis, salesperson performance, product category analysis and order-status tracking.

This project demonstrates practical skills in Excel-based reporting, data preparation, KPI calculations and business data visualization.

2. PROJECT OBJECTIVES

The main objectives of this project are:

- To inspect and understand raw sales data.
- To clean and standardize inconsistent data.
- To identify duplicate records and invalid values.
- To calculate sales, discounts, estimated costs and profit.
- To calculate key performance indicators (KPIs).
- To analyze sales performance across months, product categories and salespeople.
- To develop a dashboard for management reporting.
- To demonstrate practical Excel skills relevant to MIS Executive and Data Analyst roles.

3. TOOLS AND TECHNOLOGIES USED

Microsoft Excel

Excel is the primary tool used for data cleaning, calculations, analysis and dashboard development.

Excel Formulas and Functions

The project demonstrates formula-based transformations and calculations using functions such as:

- TRIM
- PROPER
- IF
- IFERROR
- SUMIFS

- COUNTIF
- VALUE
- DATEVALUE
- EDATE

These functions support text standardization, conditional calculations, error handling, date conversion and data aggregation.

Data Analysis Techniques

- Data cleaning and standardization
- Data validation
- Duplicate identification
- KPI calculations
- Monthly trend analysis
- Category-wise performance analysis
- Salesperson performance comparison
- Dashboard reporting

4. DATASET DESCRIPTION

The project uses a sample sales dataset containing 650 sales records and 12 original data fields.

The dataset represents sales transactions with information related to orders, products, customers, salespeople and transaction values.

The raw data includes examples of inconsistent capitalization, extra spaces, alternative city spellings, missing values, mixed date formats and invalid numeric entries. These characteristics provide opportunities to demonstrate data cleaning and validation techniques.

The cleaned dataset contains additional standardized and calculated fields to support analysis and reporting.

Note: The dataset is intended for portfolio demonstration and does not represent confidential information from an actual company.

5. PROJECT WORKFLOW

The project follows a structured data analytics workflow.

Step 1: Raw Data Inspection

The original dataset was reviewed to understand its structure, identify available fields and detect potential data-quality issues.

Step 2: Data Cleaning

Text values were standardized, unnecessary spaces were addressed, and city names were normalized using mapping logic. Date and numeric fields were prepared for analysis.

Step 3: Data Validation

Records were checked for duplicate order identifiers, invalid quantities and other potential inconsistencies. Records requiring investigation should be reviewed before using the results for business decisions.

Step 4: Calculated Fields

Additional fields were used to calculate sales values, discounts, net sales, estimated costs, estimated profit and profit margin.

Step 5: KPI Calculation

Key metrics were calculated to summarize overall sales performance and support management reporting.

Step 6: Dashboard Development

The results were organized into a dashboard with KPI summaries, charts and performance breakdowns.

Step 7: Reporting and Interpretation

The dashboard provides a structured view of sales trends and performance across different business dimensions.

6. DATA CLEANING AND VALIDATION

Data cleaning is an important stage because inaccurate or inconsistent data can lead to misleading reports.

The following data-quality areas are addressed in the project:

Text Standardization

Functions such as TRIM and PROPER help remove unnecessary spaces and standardize text formatting.

City Name Standardization

Alternative spellings and formatting variations are addressed through standardization logic. A reference mapping table can improve consistency and maintainability.

Date Handling

Date fields are prepared for time-based analysis. Mixed date formats should be checked carefully to avoid incorrect date interpretation.

Numeric Validation

Quantities, prices and discounts should be reviewed for missing, zero, negative or invalid values.

Duplicate Identification

Duplicate order identifiers are flagged for review. A repeated identifier should not automatically be deleted because the records may require business-context verification.

Data Quality Review

Records flagged by validation rules should be investigated before the final report is used for decision-making.

7. KEY PERFORMANCE INDICATORS (KPIs)

The dashboard focuses on the following business metrics.

7.1 Total Orders

Represents the number of orders included in the reporting dataset. The treatment of duplicate identifiers should be defined consistently.

7.2 Gross Sales

Represents the sales value before discounts, based on the workbook's calculation logic.

7.3 Net Sales

Net sales are calculated by subtracting the discount amount from gross sales.

Formula:

> **Net Sales = Gross Sales – Discount Amount**

7.4 Estimated Profit

Estimated profit is calculated by subtracting estimated costs from net sales.

Formula:

> **Estimated Profit = Net Sales – Estimated Cost**

The estimated cost is based on illustrative assumptions and should not be interpreted as verified business expenditure.

7.5 Profit Margin

Profit margin measures estimated profit as a proportion of net sales.

Formula:

> **Profit Margin = (Estimated Profit ÷ Net Sales) × 100**

7.6 Average Order Value

Average Order Value (AOV) represents the average net sales value per counted order.

Formula:

> **Average Order Value = Net Sales ÷ Total Orders**

7.7 Monthly Sales Performance

Monthly sales analysis helps track changes in net sales and estimated profit over time.

7.8 Category-Wise Sales

This metric compares sales across product categories and helps identify categories contributing to overall sales.

7.9 Salesperson Performance

Salesperson analysis compares sales, estimated profit and profit margin to understand differences in performance.

7.10 Order Status Analysis

Order status reporting summarizes the recorded distribution of orders across available statuses, such as delivered, pending, cancelled or returned, where those categories exist in the dataset.

8. DASHBOARD OVERVIEW

The Excel dashboard presents a consolidated view of sales performance and management reporting metrics.

The dashboard is intended to support the following activities:

- Monitoring overall sales performance.
- Reviewing monthly sales and profit trends.
- Comparing product category contributions.
- Evaluating salesperson performance.
- Reviewing order-status distribution.
- Identifying areas requiring further investigation.

The dashboard combines summary metrics with charts and supporting calculations to make the information easier to interpret.

Insert Screenshot Here

Figure 1: Sales Performance & MIS Dashboard developed using Microsoft Excel.

9. BUSINESS VALUE

The project demonstrates how an Excel-based MIS dashboard can support business reporting and analysis.

Improved Reporting

Structured reporting makes it easier to review important business metrics in one place.

Better Data Quality

Cleaning and validation checks help identify inconsistencies before reports are prepared.

Performance Monitoring

Monthly, category-wise and salesperson-wise summaries provide different views of business performance.

Decision Support

Well-defined KPIs help managers understand sales activity and identify areas that may require attention.

Reporting Efficiency

Reusable formulas and structured worksheets can simplify recurring reporting tasks when the source data and calculation logic remain consistent.

10. BUSINESS INSIGHTS

The dashboard provides a framework for identifying business insights from sales data.

After verifying the recalculated dashboard values, the following questions can be investigated:

- Which product category contributes the most to net sales?
- Which month records the highest net sales?
- How does estimated profit change over time?
- Which salespeople have the highest net sales?
- Are there differences between sales performance and estimated profit margin?
- What proportion of recorded orders falls into each order-status category?
- Which data-quality issues should be resolved before management reporting?

Important: Specific findings should be added only after reviewing the actual calculated values in Excel. No numerical business conclusions are assumed in this report.

11. LIMITATIONS AND ASSUMPTIONS

The project has the following limitations:

- The dataset is sample data and does not represent verified company transactions.
- Estimated costs use illustrative assumptions rather than actual financial records.
- Profit and profit margin therefore represent estimates, not audited financial results.
- Duplicate order identifiers and invalid values require review before final reporting.
- Automated standardization may require manual verification for ambiguous entries.
- Dashboard results depend on the accuracy of formulas and successful recalculation in Excel.

For real business use, cost assumptions should be approved by the finance team, and the treatment of returns, cancellations, discounts and duplicate transactions should be clearly defined.

12. SKILLS DEMONSTRATED

This project demonstrates the following skills:

- Microsoft Excel
- Data cleaning and transformation
- Data validation and quality checks
- Excel formulas and functions
- KPI calculation
- Sales performance analysis
- Profitability analysis using assumptions
- Trend analysis
- Dashboard development
- MIS reporting
- Business-oriented data interpretation
- Documentation and reporting

These skills are relevant to entry-level MIS Executive and Data Analyst positions.

13. FUTURE ENHANCEMENTS

The project can be enhanced further by:

- Adding interactive Excel slicers and filters.
- Creating a formal data dictionary.
- Improving duplicate and invalid-record reporting.
- Using a reference table for city-name standardization.
- Automating data preparation with Power Query.
- Developing an interactive Power BI dashboard.
- Adding verified business insights and recommendations.
- Creating a refreshable monthly MIS reporting process.
- Adding a separate data-quality summary for management review.

14. CONCLUSION

The Sales Performance & MIS Dashboard demonstrates an end-to-end Excel reporting workflow, from raw data inspection and cleaning to KPI calculations and dashboard presentation.

The project highlights the importance of data quality, consistent metric definitions and clear reporting in business decision-making.

By combining Excel formulas, data validation, sales analysis and visualization, the project provides a practical portfolio example relevant to MIS Executive and Data Analyst roles.

Further enhancements, including automated data preparation, interactive Power BI reporting and verified business insights, can expand the project's analytical capabilities.

Prepared by: Sandhya Shree G M

Project: Sales Performance & MIS Dashboard

Primary Tool: Microsoft Excel

Portfolio Purpose: Demonstration of Excel-based MIS reporting and data analytics skills